Design and Fabrication Methods for Hemp-Based Acoustic Elements

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This study investigates and proposes design and fabrication methods for hemp-based heat-pressed acoustic tiles. Through computational pattern generation and acoustic analysis methods, combined with CNC-milling of formwork for heat pressing and room acoustic measurements are material-geometric relations examined and described. The study results in new procedural design method for creating bespoke visual-acoustic modules and knowledge of how these hemp-based tiles can impact the acoustic environment.
Keywords: *Hemp, Acoustics, Pattern Generation, Heat Form Pressing*

INTRODUCTION

Acoustics have a direct and fundamental impact on our ability to understand an environment through the reverberation of sound energy within an enclosed space. The spatial response to sound energy takes part in defining how humans decode and use space in varied conditions and during different activities (Long, 2014; Rasmussen, 1964; Zumthor, 2012). As the predominant amount of sound energy arriving at the human ear has been reflected from a nearby surface, the material-geometric understanding, composition and making of such surfaces are central in defining human acoustic sensations. The study, design and making of architectural acoustic surfaces have been studied in a variety of ways, through complex computational and analytical design structures (Cox & D'Antonio, 2016; Long, 2014; Reinhardt et al., 2012) and their making processes by advanced fabrication technologies (Foged et al., 2022, 2014; Peters & Olesen, 2010; Rossi et al., 2021; Walker & Foged, 2018). Many investigations utilize modular configurations to define sound absorption and sound scattering. The varied modular geometries enable greater surface area than a planar surface, which in turn provides the basis for systematically steering the sound energy absorption and reflection response of the surface developed. Investigation and testing of modular acoustic elements, with varied material-geometric configurations is the background for the present study. The geometric modularity presented in the previous studies include, however, also the caveat of complex and often costly fabrication processes through robotic processing and use of materials with high embodied energy (such as ceramic tiles, steel plates, plaster elements). The background on potential solutions in the presented study is focused on the use of an industrially available recycled hemp batten that is heat pressed into a bespoke, modular, and reconfigurable acoustic tile geometry. The potential for new solutions is attempted through experimental design studies with computational pattern generation methods, with integrated acoustic analysis methods, room acoustic measurements and design exploration of the developed methods and models. The paper presents the developed methods, models, a full- scale demonstrator, the results obtained through the studies and a discussion on the results.



Figure 1
Hemp-based material in uncompressed form, compressed form (A side of formwork) and compressed (B side of formwork. All images in 10x magnification.

METHODS

The project is based on a mix-method experiment driven approach, including material studies of form-pressed hemp battens with a Polythylene (PE) binder, computational design pattern-generation, computational acoustic analysis, prototype experiments and room acoustic measurements.

Material Studies

Hemp, formed as battens with 20mm thickness (25 kg/m^3), including PE binder is investigated as a transformable material through heat induced form-pressing. By pressing the composite material under heat, the binder is melted while the hemp fibers are moved closer together. Depending on pressure added, the process transforms the material characteristics from an open porous structure to a closed dense structure. The material transformation changes at the same time the structural and acoustic qualities of the material, making the material stiffer and less acoustically absorbing, especially at higher frequencies, as other porous absorbers (Long, 2014). A bespoke pressing tool is developed and fabricated through CNC milling of two aluminum blocks (400x400mm), into two positive forms, mounted in a bespoke press. The form tool geometry is developed through below described Pattern Generation Studies. Both the aluminum forms and the hemp battens are heated to 160 degrees Celsius for 10 minutes before they are mounted in the press, figure 2. The pressure exerted to the hemp batten is 250 bar, kept for 10 mins. under pressure. Values for temperature, duration, pressure and pressing process is found under trial-error experimental processes with the aim to understand and identify how tight angled and local form-changes can be made without breaking the hemp fibers, which results in material-form failure, as seen in figure 1.

Computational Pattern Generation

Investigation and development of form and material properties for acoustic articulations are based the generative pattern software Parakeet for Rhino-Grasshopper (Mottaghi & Beigi Khalil, 2022). Patterns are explored in 2D variations, resulting in 3D emboss effects of the material. This process induces local densities across the porous hemp batten by creating elevated forms where the 2D pattern lines are drawn. Computational pattern explorations are delineated by a base grid of 400x400mm (maximum possible press field), combined with two opposing arch forms, which again have multiple parallel lines. The number of offset arches, and the rotation of the arches within the base grid defines the pattern variables for versioning studies. The reduced set of pattern variables enables a single form-tool, ones the pattern has been developed for full scale prototyping and acoustic testing. This approach addresses a common issue of complicated, expensive fabrication processes of curved, hybrid acoustic surfaces (Cox & D'Antonio, 2016) and enables a potential fast, and cost-effective manufacturing process compared to additive manufacturing, casting, injection-molding or CNC milling of complex surfaces.





Figure 2
Bespoke press with aluminium formwork (top) and thermography showing the heated hemp and aluminium before pressing (bottom).

Computational Acoustic Analysis

Acoustic analysis is implemented directly into the computational design model by programming a RhinoGrasshopper component that calculates the room acoustic reverberation time (T60) based on the Sabine and Norris-Eyring formula. The Sabine formula is considered accurate for spaces with absorption levels < 0.5, by use of (Long, 2014).

$$RT60 = \frac{0.161\,V}{At} \qquad (1)$$

Where V is the space volume and At is the total absorption of the space in m^2. The physical testing space used is also used in the computational design investigations. The geometry of the space is a shoebox geometry, figure 3, with 7.6m, 5.6 m, 2.8m in width, depth, height. The surfaces are hard, with large glass windows, plaster walls and ceiling, and linoleum flooring. Textile curtains are placed in the corners of the space, retracted in both the computational model and during room acoustic measurements. While the space is considered a low absorption space, the insertion of the hemp battens will create a more complex material-space condition. In order to explore and analyse the room acoustic impact from the patterned hemp elements, the Norris-Eyring acoustic analysis method is used. By calculating the specific surface to the specific absorption coefficient in the space a more accurate material-geometry is obtained, allowing assessment of spaces with alpha < 0.2 (Long, 2014). The Norris-

Eyring formula is based on an extention of the Sabine and Millington-Sette formula.

$$RT60 = \frac{0.161\,V}{-St * ln(1-\alpha)} \qquad (2)$$

Where V is the space volume, S_t is the total surface area, αw is the weighted absorption coefficient, which again is defined by:

$$\alpha w = \frac{S1\alpha 1 + S2\alpha 2 + \cdots + Sn\alpha n}{St} \qquad (3)$$

In a geometrically simple space, as the shoebox geometry without exposed columns, beams and shadow areas, the methods are considered both accurate and effective for design computation. During computational design experimentation studies, each iterative pattern versioning and acoustic room analysis iteration was conducted within 1 second on a laptop machine, making large explorative search procedures feasible compared to more computational expensive procedures. The absorption coefficient of the developed hemp composite, used for calculating the weighted absorption of the space, is defined by the specific material characteristics, including the bulk density (kg/m^3), material thickness (mm) and flow resistivity (Pas/m^2). Tabular data of porous materials (polyester fibers) flow resistivity can be found in the literature (Garai & Pompoli, 2005) or through calculation (Modeling, 2022). An almost linear relation exists between the material bulk density and the flow resistivity in porous materials. The hemp material is press-formed into three material conditions, *thickness* = 40mm (untreated), 15mm (surfaces) and 5mm (folds), *densities* = 85 kg/m^3, 170 kg/m^3 and 680 kg/m^3, with estimated *flow resistivity* = 10000 Pas/m^2, 20000 Pas/m^2 and 60000 Pas/m^2. Using (Modeling, 2022), we find the sound absorption coefficients for each material state across the octave bands. The data is summarized in table 1.

		Hemp A	Hemp B	Hemp C
Density	kg/m^3	25	75	225
Thickness	mm	40	15	5
Flow res.	Pas/m^2	2700	8800	10000
α	125 Hz	0.05	0.02	0.01
α	250 Hz	0.15	0.03	0.01
α	500 Hz	0.22	0.1	0.02
α	1 kHz	0.38	0.15	0.04
α	2 kHz	0.65	0.28	0.08
α	4 kHz	0.63	0.55	0.11

Table 1
Material-acoustic data used for simulating RT60 across 125Hz-4kHz.

Following the values in table 1, combined with the integration of the Norris-Eyring formula with the pattern design computation model, it is possible to acoustically evaluate each pattern generated during design experimentation.

Room Acoustic Measurements

Room acoustic analysis is based on the developed material-formwork, organized in rectangular modules assembled into an 8 m^2 surface structure mounted on a 9mm MDF plate, positioned against one wall in the test space, figure 3. Room acoustic analysis is conducted with a calibrated Behringer 8000 measurement microphone, a semi-directional Yamaha MSP5 loudspeaker and Fuzzmeasure by Røde room acoustic software. Measurements follows ISO procedures (ISO, 2008), using 8 receiver measurement points and 2 source points. 4 receiver



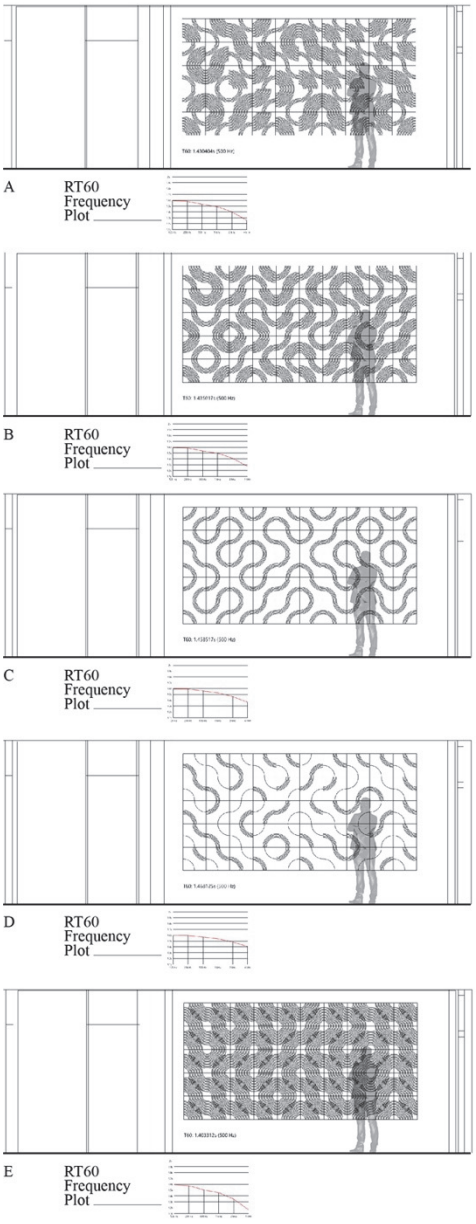
Figure 3
Room acoustic measurements with full scale demonstrator.

Figure 4
Geometric-acoustic iterative output of the developed computational design method and model.

points are in standing position height and 4 in seated position height. The room temperature and relative humidity during measurements were measured to 20 degrees Celsius and 53 %, respectively. Test measurements were conducted to determine sweep impulse duration, set to 2 seconds following calibration. The impulse response was analysed and computed for Reverberation Time (T30), Definition (D50) and Clarity (C80). Loudness and Sound Pressure Levels where computed but not included here due to the small test space with little sound pressure drop.

DESIGN EXPERIMENTATION

Based on the developed computational design model, combining parameterized pattern generation and room acoustic analysis, examination of the method and model is done through an experimental design process. The objective is not to propose a specific and optimized acoustic condition and composition, but rather to explore the method and model's ability to visually and acoustically generate, analyse and present possible solutions that are informed by material, geometric and acoustic factors. By the change of three model input factors, 1) rotation of the square base tile, 2) first group of consecutive arch geometries and 3) second group of consecutive arch geometries, a large set of visual output compositions and acoustic absorption characteristics are identified, figure 4. Each version of the experimental run includes the geometric articulation (which is translated to the milling pattern for the press tool used to form the hemp tiles), of the room acoustic performance, graphed as a RT60 plot across 125Hz, 250hz, 500Hz, 1kHz, 2kHz, 4kHz octave bands, and the collective catalogue of solutions outputted in a matrix allowing a direct and comparative analysis for the designer, figure 5-6.



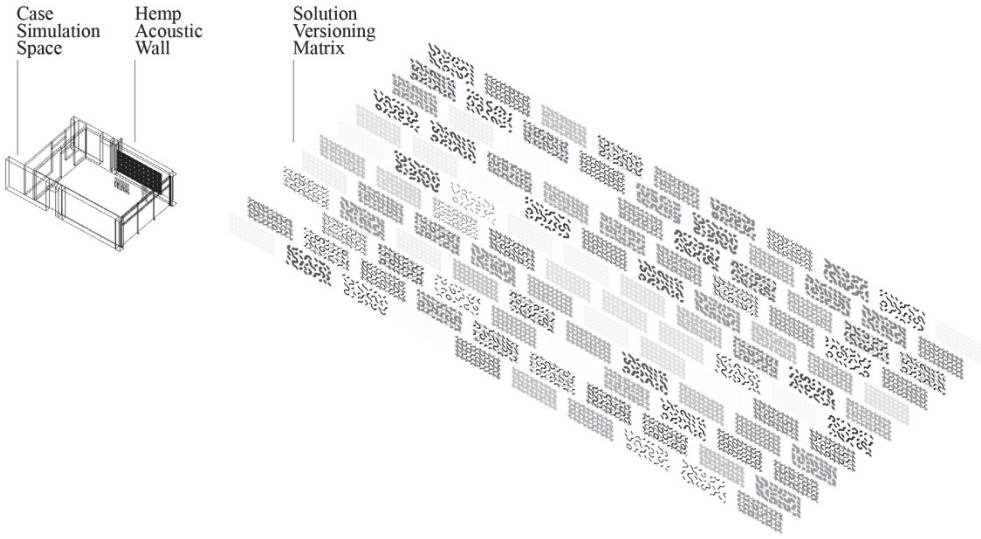


Figure 5
Simulation space, latest computed material-geometric composition and solution matrix mapping for comparative evaluation.

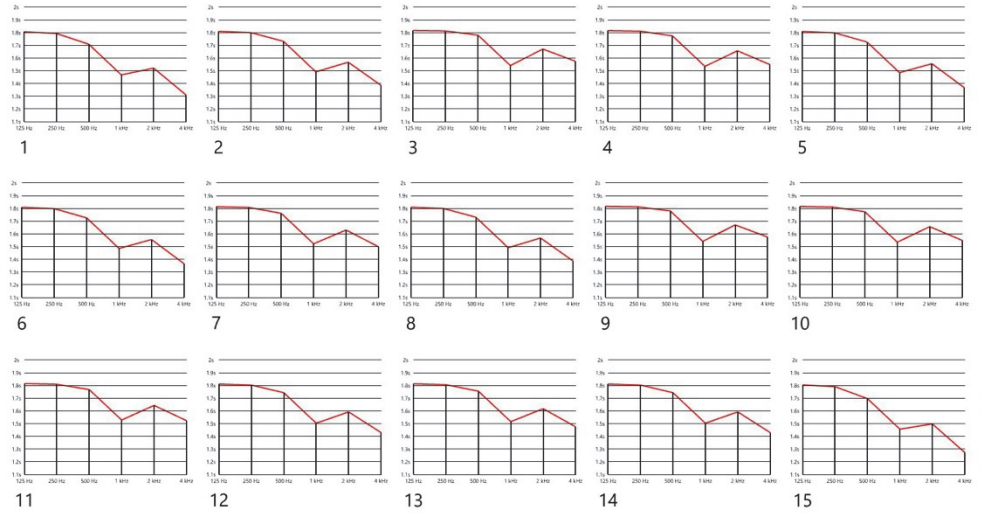


Figure 6
Reverberation Time (RT60) graphs output for each design iteration (above).

RESULTS

The study investigates how an industrial, readily available product of hemp battens, can be designed and fabricated through computational pattern generation, CNC-manufacturing and heat press methods for specific acoustic absorption and room acoustic properties. The study illustrates how the hemp modular elements are processed by pressing a developed pattern into the material through bespoke formwork, which induce new properties in the material by sub-element specific strength and stiffness characteristics with associated change in material densities that alter the sound energy absorption properties of the original porous material. The study finds that the new method of combining computational pattern generation with a room acoustic analysis method enables a fast and explorative design process, where a design conjecture can be informed by the computational generation of solutions, developed in seconds on a laptop computer, and subsequently supporting design evaluation through multiple design objectives (acoustics, visual, structural, material use, formwork fabrication). The room acoustic measurement studies document the impact of the developed demonstrator, where the RT60 has been reduced from an average of 1.3s (250Hz-4kHz) to 1.0s (250Hz-4kHz) with the case aim to construct a room acoustic condition useful for seminar and lectures rooms.

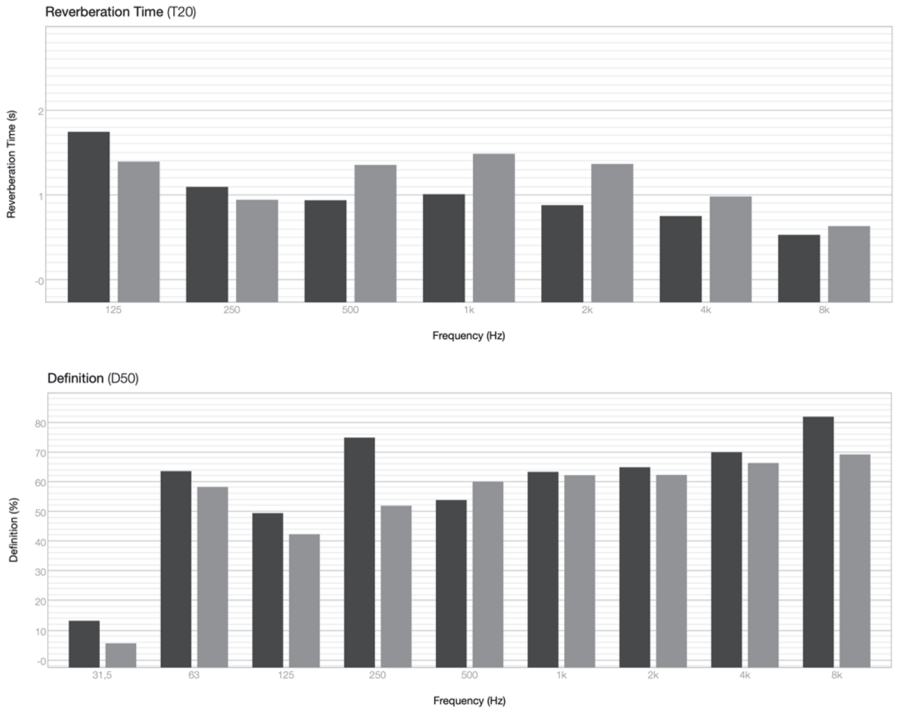


Figure 7
Room acoustic measurement results with demonstrator (dark grey) and without demonstrator (light grey) in the space.

DISCUSSION

1. The process of pressing the hemp Bi-Co composite panels was subject to slight deviations in temperature and pressure, as the hydraulic pump was hand controlled by different operators. The aluminium tool was re-heated for every third pressing interval, thus inconsistent in temperature. Ideally the tool should have an active heating system and set to an operating temperature of 150 degree Celsius.
2. The three-dimensional deformation of the hemp fibres by the pressing tool (positive arch pattern) bordered limits of the fibrous material. Transitions between height and depth proofed to be too abrupt, causing a) the high areas not to reach the intended level of deformation with a resulting surface area smaller than intended, thus likely to be less effective in scattering sound energy; b) breakage / tearing of hemp fibres (visually).

3. A more general aspect of discussion rests in a superior sound absorption performance of uncompressed (2x20mm) low density hemp battens, compared to compressed material with increased densities in the main areas of 15mm and especially within the circle pattern area, that is compressed to a section thickness of 5mm. The argument of embossing and compressing the composite material, is placed in an intended balance between material performance (acoustic damping abilities) and structural soundness (rigidity and exposable surface physiognomies) by intentional incorporating variable and tile able patterns with high- and low densities.

4. The material used for prototyping, is commercially available and acquired from Convert A/S. The battens are composed in a non-woven structure of 70% hemp fibres and 30% binder in form of Polyethylene (PE) Bi-Co fibres. This irreversible combination of materials is questionable in terms of re-usability and bio-degradation. In future iterations it will be interesting to replace the PE fibres with a bio-degradable polymer fibre, such as a low-melt temperature PLA (Polylactic Acid), to reason for clean compostable end-of-life scenario.



Figure 8
Full scale demonstrator as installed in the measurement space composed of 45 identical tiles.

CONCLUSION

The research objective was to study how an airy non-woven hemp-PE BiCo fibre composite structure in conjunction with external forming can be developed for specified acoustic-structural properties through heat-pressing. The aim extends to studying how the elements can be developed to perform structurally as to avoid a secondary load-bearing element, while retaining sound absorption qualities as a porous medium.

Three criteria were investigated and concluded upon.

1. The hemp-composite materials heat forming/compression abilities by utilizing a preheated aluminium mould, observing a good surface finish of the hemp panels, a good release (does not stick) ability and a fairly even heat distribution in the aluminium block. The hemp-composite is able to obtain and retain three-dimensional double curved forming. Extreme steep shifts are challenging and cause shearing and breaking of fibres. Concluding to allow for larger transition area at deformation, in future iterations.

2. New understanding of the compressed hemp fibre's ability of damping sound energy has been created, thus enabling predictability in a variable future cases and implementation in computational design systems.

3. Establishing a design chain between computational methods and systems has been documented, which in turn allow for future studies and variations in modular patterns. The bespoke pattern generator with acoustic analysis is central in this study, enabling predictability of the room acoustics, which vice versa acts as design criteria for potential generative design iterations that employ search algorithms. The specific demonstrator is based on embossing and compressing a flat panel (which focused on affordability in respect to cost and time, using only one CNC-milled tool part to create complex visual-acoustic properties). In future studies, it will be interesting to investigate the pressing of panels into deeper, spatial shell structures with anisotropic densities to explore a more articulated appearance and to enhance sound scattering and absorption abilities of the modular panel structure.

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